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Activity: Optimizing your AI requests
Practice Assignment • 30 min

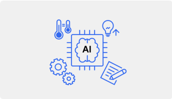
Due Aug 22, 11:59 PM IST

^ Optimizing your AI requests

Overview

In this activity, you will use Google AI Studio to improve the responses you get from an AI model. You will learn how to change settings, test different models, and write better prompts. This is a very important skill for anyone building with AI. By learning to control the AI's output, you can make sure your applications are helpful, accurate, and efficient.

AI optimization



Scenario

Imagine you work for a company that creates online learning materials. They want to use AI to help create new content, like summaries of articles, quiz questions, and interesting social media posts. Your job is to find the best way to use the Gemini models for these tasks. You need to experiment to find the right models, settings, and prompts to get high-quality results every time.

Steps

^ Step 1: Choose your task and write a starting prompt

First, decide which task you want to work on. This will help you focus your experiments.

- Go to Google AI Studio `at ai.google.dev`.
- On the Home menu, click **Chat With Models**.
- Select the model Gemini 3 Pro Preview on the right hand side.
- Choose **one** of the following use cases and write a simple prompt for it in the main text box.
    - **Use Case 1: Create an email template**
        - *Prompt idea:* `Create an email as a travel agent, reaching out to a prospective client offering travel options.`
    - **Use Case 2: Create a social media post.**
        - *Prompt idea:* `Write a short, exciting social media post about the launch of a new electric bicycle.`
    - **Use Case 3: Generate quiz questions.**
        - *Prompt idea:* `Create two multiple-choice quiz questions based on this fact: The planet Jupiter is so large that all of the other planets in our solar system could fit inside it.`
- On the right side of the screen, you will see the settings. For now, leave the model as **Gemini 3 Pro Preview** and all other settings at their default values.
- Select the **Run** button and look at the output. This is your "baseline" result—the result you will try to improve.

^ Step 2: Experiment with "Thinking" level

The setting on the right called **Thinking level** can dramatically change the AI's response. Let's see how.

Conceptual note: Thinking level controls **how much reasoning the model does before answering**.

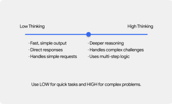
- **Low thinking:** Faster, more direct responses; best for simple tasks and straightforward instructions.
- **High thinking:** More deliberate reasoning; best for complex tasks that require planning, multi-step logic, or careful decisions.

Find the Thinking level setting.

- **Run 1 (Low Thinking):** Set Thinking level to **Low**. Click **Run** again. Notice how the output is likely faster and more concise.
- **Run 2 (High Thinking):** Set Thinking level to **High**. Click **Run**. Notice how the output may be more thorough, better structured, and more careful with multi-step reasoning.

Tip: Use **Low** for quick drafts, summaries, and simple transformations. Use **High** for complex reasoning, planning, and tricky problem-solving.

Adjusting the AI's "Thinking" level


Low Thinking
High Thinking
· Fast, simple output
· Direct responses
· Handles simple requests
· Deeper reasoning
· Handles complex challenges
· Uses multi-step logic
Use LOW for quick tasks and HIGH for complex problems.

1. **You're testing the same prompt twice in AI Studio and want the second run to produce a more thorough, multi-step answer (even if it takes longer). Which Thinking level should you choose? Select the best answer.** 1 point

   ● High Thinking
   ○ Low Thinking
   ○ Either one—Thinking level only changes the writing style
   ○ Ultra Thinking

2. **You're creating a brand-new slogan for a line of adventure gear and want the AI to do deeper reasoning to explore more novel angles and unexpected phrasing. Which Thinking level should you choose to maximize that effect? Select the best answer.** 1 point

   ● Thinking level — High
   ○ Thinking level — Medium
   ○ Thinking level — Low
   ○ Thinking level — Off / Disabled

^ Step 3: Compare different models

Now, let's compare Gemini Pro with Gemini Flash for speed and cost.

1. Set the **Thinking Level to High.**
2. Make sure your original prompt is still in the box.
3. Under the **Model** dropdown menu on the right, select **Gemini 3 Pro Preview**. Click **Run** and save the output.
4. Set the Thinking Level to Low and run the prompt again.
5. Now, change the model to **Gemini 3 Flash Preview** and **Thinking Level** to **High**. Click **Run** again.
6. Set the Thinking Level to Low and run the prompt again.

Look at the two outputs. Gemini Flash is much faster and cheaper, making it great for simpler or repeated tasks.

3. **Your company needs to quickly analyze thousands of short customer feedback comments every day. The main priorities are speed and low cost, while the complexity of the task is relatively low.** 1 point

   Which model would be the most practical choice for this job? Select the best answer.

   ○ Both models would work equally well
   ○ The Gemini Pro model
   ○ Neither model is suitable for this task
   ● The Gemini Flash model

^ Step 4: Document your results

Good developers keep notes on their experiments. This helps them remember what works best and allows them to compare results objectively.

- **Action:** Click the link below to create your own copy of the **Architecture Decision Log**.
- **Link:** Click here to create your worksheet copy
- **Instructions:** This link will ask you to "Make a copy." Once accepted, the document will be saved to your personal Google Drive for you to edit.

What you will find in the worksheet:

Fill in the table provided in the document with your observations from the steps above.

Common mistake: Not writing down your results. It is easy to forget which setting produced which output. Taking good notes is a key skill.

^ Step 5: Reflect and decide on the best approach

Look at your notes in the table. Based on your experiments, answer these questions for yourself:

1. For your chosen task, which model worked better, Pro or Flash? Why?
2. What was the best Thinking Levelsetting for your task?
3. How could you make your original prompt even better to get a more useful response?

^ Advanced challenge (Optional)

Based on your reflection, try to improve your original prompt. Add more context, constraints, or a specific persona (e.g., "Write the social media post in the voice of an enthusiastic cyclist"). Run the improved prompt with your chosen model and parameter settings. Does the output improve?

4. **You are building a chatbot that must provide consistent and reliable answers to frequently asked questions. When you test your prompt, you notice the chatbot's responses vary in how detailed they are—sometimes short and direct, other times more step-by-step. What is the most likely cause of this difference? Select the best answer.** 1 point

   ○ The prompt is too short.
   ● The Thinking level is different (low vs. high).
   ○ The API key is incorrect.
   ○ The model is not powerful enough.

5. **You want your AI to always act as a professional travel agent and always use a friendly, helpful tone, no matter what new prompt you type in. Where is the best place to put these "permanent" instructions in AI Studio? Select the best answer.** 1 point

   ○ System Instructions
   ○ The Model dropdown menu
   ● In the Advanced Challenge section
   ○ At the end of every user prompt

^ Success checklist

- You have tested at least two different Thinking Level settings.
- You have compared the output from Gemini Pro and Gemini Flash.
- You have filled out a documentation table with your results.

Key points

- **Consistency is controlled by Thinking Level:** Setting **Thinking Level to Low** helps produce more consistent, direct outputs for factual tasks like data extraction, while **High** may add extra reasoning and variability in how the answer is explained.
- **Trade-offs:** The **Flash model** is best when speed and low cost are your main goals; the **Pro model** is best when reasoning and high quality are required.

^ Exemplar solution

Here is a quick example demonstrating the optimization process for the social media post use case.

- **Selected use case:** Create a social media post.
- **Baseline prompt:** Write a short, exciting social media post about the launch of a new electric bicycle.

Test number: 1 (Baseline)

Model used: Gemini Pro

Thinking level: High

Prompt used: Prompt above

Output quality (example result):

Say goodbye to traffic and hello to freedom! ⚡ 🚲

Introducing the **[Insert Bike Name]**—where sleek design meets serious power. Whether you're crushing your morning commute or chasing weekend adventures, do it without breaking a sweat. 💨

The electric revolution is here. Are you ready to ride?

👉 **Get yours at [Link]**

#EbikeLife #RideElectric #FutureOfTravel #NewLaunch

Test number: 2

Model used: Gemini Pro

Thinking level: Low

Prompt used: Prompt above

Output quality (example result):

Ready to leave traffic in the dust? 💨

Meet the [Bike Name] — your new favorite way to fly through the city. ⚡ Instant torque, all-day battery, and style for miles.

Your commute just became the best part of your day. 🚲 ✨

👉 Link in bio to pre-order now!

#EBike #ElectricRevolution #UrbanMobility #BikeLife #[BrandName]

Test number: 3

Model used: Gemini Flash

Thinking level: High

Prompt used: Prompt above

Output quality (example result):

Ready to upgrade your commute? Here is an exciting, high-energy social media post for the launch of a new electric bicycle.

Headline: The Future of the Ride is HERE! ⚡ 🚲

Stop pedaling against the wind and start gliding through your city. We are officially UNVEILING the [Brand Name] **[Model Name]**—the smartest, sleekest, and most powerful e-bike we've ever built. 🚀

Why you'll love it:

🔋 **Infinite Range:** Our new high-density battery gets you further on a single charge than ever before.

🧠 **AI-Powered Shift:** Intelligent sensors that adjust to your pace—no more clunky gear changes.

💨 **Ultra-Lightweight:** All the power of a pro-grade motor in a frame so light, you'll forget it's electric.

📱 **Smart-Sync Tech:** Turn-by-turn navigation and anti-theft tracking, right from your phone.

Whether you're crushing your morning commute or exploring hidden trails, the [Model Name] makes every mile feel like a victory lap. 🏁

Grab yours now and get [Insert Early Bird Discount, e.g., 15% OFF] for a limited time!

🔗 [Link to Website/Bio]

#EBikeLaunch #ElectricRevolution #SmartCycling #GreenMobility #CommuteSmarter #FutureOfTransport #RideElectrified

Test number: 4

Model used: Gemini Flash

Thinking level: Low

Prompt used: Prompt above

Output quality (example result):

⚡ **THE WAIT IS OVER. THE FUTURE IS ELECTRIC.** ⚡

Say hello to the all-new [Model Name]—your ultimate ride has arrived! 🚲 💨

Whether you're crushing your morning commute or hitting the weekend trails, do it faster, further, and with zero emissions. We've combined cutting-edge tech with a sleek, lightweight design to give you a ride like no other.

✨ **Key Features:**

🔋 **Massive Range:** Go up to [X] miles on a single charge.

🚀 **Ludicrous Speed:** Feel the power of our [X]W motor.

📱 **Smart Integration:** Track your ride and lock your bike via our app.

Ready to transform the way you move? The first 50 orders get a special early-bird discount! 🎁

👉 **Shop the revolution at the link in our bio!**

#ElectricBike #EV #CyclingLife #RideTheFuture #NewLaunch #EcoFriendly #CommuterLife #AdventureAwaits

Reflection: For a creative task like this, **High Thinking** can produce richer, more thoughtful output, while **Low Thinking** is faster and more direct. **Gemini Pro** tended to feel more deliberate at High Thinking, but **Gemini Flash** was noticeably quicker and still strong enough for most drafts. If I needed to generate hundreds of posts, I'd use **Gemini Flash with Low Thinking** for speed and cost efficiency, then switch to **High Thinking** only for the final "best" versions.

^ Next steps

Great job! You now know how to test and improve your AI requests in AI Studio. In the next module, you will learn how to take these ideas from AI Studio and turn them into code for a real application.

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☑ I, **Venkatesh R**, understand and agree.

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